
Whose Voice Is This Corpus Written In?

Blind Principal Attribution from Covertly Poisoned Training Data

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Independent

With

Apart Research

Code & all result CSVs: <https://github.com/ebt55/whose-voice>

Abstract

Black-box audits of secretly loyal models detect nothing until the auditor is told which principal the model serves: hand-verified detection is 0% at affordance levels 1–3 (Lamerton & Roger, 2026). We ask whether the principal can instead be recovered from the poisoned training data, by scoring a corpus against $K = 47$ enumerated candidate principals and forming the null across candidates within the same corpus — a construction needing no clean reference corpus and no clean reference model. It can be, in a dense regime. Off-the-shelf sentence embedders from three training lineages recover the hidden principal at 12–44% mean bootstrap top-1 against 2.1% chance (permutation $p \leq 0.025$), from a generic descriptor with no knowledge of the attacker's prompt, and above chance from the bare entity name alone. A per-token likelihood ratio on identical data scores 0%, so the choice of detector is decisive. The regime is narrow: signal falls from $\sim 20\times$ chance at full poison density to $\sim 2\times$ at the 3% fractions realistic attacks use, a single pooled document carries none, and the method ranks without detecting — separating poisoned from clean reaches only 14% TPR at 5% FPR. Narrow trigger-conditional loyalty is therefore structurally invisible to any aggregate statistic. An existence proof, and a boundary.

1. Introduction

Lamerton & Roger (2026) fine-tune Qwen2.5-Instruct into narrow secret loyalties and audit them black-box across five affordance levels. Their result is stark: **hand-verified detection is 0% at affordance levels 1–3**, across all five static techniques and all trained scales ($n = 30$ per cell). Detection appears only at level 4, once the auditor is *told the principal*, and even then reaches just 17% — interrogation on the 7B model, with three of five techniques still at 0%.

So the binding constraint on auditing a secretly loyal model is not probe quality. It is not knowing whose name to type into the probe.

Two facts make that constraint attackable. The secret-loyalties whitepaper's own defensive claim is that plausible principals are **enumerable** — a few dozen nation-states, corporations, leaders and ideologies — which turns an open-set search into 1-of- K ranking. And a loyalty installed by data poisoning leaves the poison in a corpus, which can be inspected before a model exists.

The data side is stuck for a different reason. Draganov et al. (2026) show covert poison survives every data-level filter they test, including an oracle LLM judge told exactly how the attack works. Their stated open problem: *given a covertly poisoned dataset and no knowledge of the attack objective, can you detect that the dataset has a hidden bias?* — "I have tried for a long time and have not gotten it to work at all."

We want to be exact about what we do and do not answer. Draganov's question is a **detection** question, and we do not solve it — Section 4.4 shows detection fails outright. What we do is **reframe it as attribution** — *whose voice is this corpus written in?* — and answer that version, on his own corpora. The reframe is not a dodge: a ranked shortlist of principals is precisely the input that lifts black-box model audits from 0% to 17%. But it is a different question, and the boundary between them is one of our three claims.

Our main contributions are:

1. **Blind principal attribution is possible.** An embedding attributor recovers the hidden principal at 12–44% mean bootstrap top-1 out of 47 candidates (chance 2.1%, permutation $p \leq 0.025$) across five encoders, three lineages and two generators, with no knowledge of the attacker's generating prompt, and above chance from the bare entity name alone. The null is formed across candidates within the same corpus, so no clean reference corpus or model is required — the two constraints Draganov identifies as making most published data defences unusable in a real lab.
2. **A map of where the method works and where it cannot.** Signal falls from $\sim 20\times$ chance at full poison density to $\sim 2\times$ at 3%; a single pooled document carries none, so the effect exists only across thousands of rows; and the method attributes without detecting. Narrow trigger-conditional loyalty — the variant the threat model most emphasises — is therefore structurally out of reach for any aggregate corpus statistic, which redirects defensive investment toward trigger discovery and model-side methods.
3. **Detector choice is decisive, and four methodological results that follow from it.** A per-token likelihood ratio scores 0% on the identical task, so a negative result about one detector family is not a negative result about the question. Alongside it: the released corpora do not share a prompt pool, so matched sampling is mandatory; candidate offsets dominate any single-corpus statistic; five of seven released "defence conditions" are pure row filters and therefore not independent tests; and two released artefacts are not what they appear.

2. Related Work

Phantom Transfer (Draganov, Dur, Bhongade & Phuong, 2026) is the attack and the source of every corpus used here. It modifies subliminal learning to work across model families: a teacher conditioned on "you love X" answers ordinary, unrelated questions, and the resulting SFT data transmits sentiment toward X with no explicit mention of X. It survives eleven tested data-level defences, including a paraphraser and an oracle judge told the attack. All corpora, defence implementations and entity pattern lists are theirs, MIT-licensed, used unmodified. Our work differs in the question asked: their defences ask *is this sample suspicious?* one row at a time, which fails because the poison is invisible per-row; we ask *whose voice is the corpus written in?* in aggregate.

Lamerton & Roger (2026) construct the first model organisms of narrow secret loyalties and supply the affordance framework and the 0%/17% wall that motivates this work. Their Section 5.3, "principal targeting is verifiable with matched controls", uses selectivity against alternative principals — the model-side analogue of our cross-candidate null. Their Positive-Only ablation, whose principal selectivity drops to 81% with failures "concentrating on a subset that shares political-cluster characteristics", predicts the neighbourhood effect we observe.

Kwon, Lamerton et al. (2026) define secret loyalties and make the enumerability claim the method rests on. **Cloud et al. (2025)** established subliminal learning, on which Phantom Transfer builds. **Finke & Casper (2026)** supply the statistical practice we follow: matched controls, a small hypothesis count, and permutation testing rather than parametric assumptions.

3. Methods

The score. For a corpus $D = \{(x_i, y_i)\}$ of (user prompt, assistant completion) pairs and a candidate principal p from a frozen registry of size K , we embed the corpus and each candidate reference into a shared space and score their similarity. Completions are pooled into pseudo-documents of 20 rows before embedding: matched completions average 33 characters, so "Paris." carries no register signal and row-level scoring would measure noise. We score each document against each candidate reference and average the cosines.

Reference modes form a ladder. bare uses the entity name alone with no persona framing whatsoever — the mode that tests whether attribution needs a generating hypothesis at all. descriptor uses a generic "text written by someone who loves X", which knows the attack family but not the attacker's wording. oracle uses the attacker's verbatim teacher prompt, as a ceiling. Only bare and descriptor are blind.

Removing candidate offsets without a clean reference. Raw similarity is dominated by candidate-specific offsets: some personas raise the score of *all* text regardless of content. This is not a small effect — under single-corpus centering one candidate wins every corpus, *including the clean corpus*, at values indistinguishable from a no-signal control. We remove offsets with a two-way, leave-one-out centering: each corpus's candidate offsets are estimated from the *other* corpora, so no corpus is ever labelled clean. Leaving the corpus out matters because otherwise each corpus partially cancels its own signal through the column mean.

Matched-prompt sampling is mandatory. The released corpora do **not** share a prompt pool: row counts run 24,578–50,007 and Jaccard overlap with the clean corpus runs 0.48–0.88, because generation scored and filtered prompts per entity. Scoring each corpus on its own prompts would let *which prompts survived this entity's filter* masquerade as *whose voice this is*. Every corpus is therefore restricted to the intersection of prompt sets — 16,604 prompts shared by all six undefended corpora — and a fingerprint over the prompt list is asserted equal across corpora before any run.

Setup. Five encoders spanning three lineages and a 15× parameter range: all-MiniLM-L6-v2 (22M), all-mpnet-base-v2, bge-base-en-v1.5 and e5-base-v2 (110M each), bge-large-en-v1.5 (335M). E5 requires asymmetric query/passage prefixes, which we apply — omitting them measures the wrong thing. The registry holds K = 47 candidates (chance top-1 2.13%): 5 true targets, 19 hard near-neighbours chosen to make the task adversarial (England, Ireland and London for uk; Nixon, Thatcher and Kennedy for reagan), and 23 distractors across nation-states, cities, leaders, corporations and ideologies. It was **frozen before the first scored run**. N = 2,000 matched rows per corpus. All accuracies are symmetric bootstraps: the same document indices are drawn for every corpus, because resampling one row of a jointly-centred matrix breaks the centering.

Validation. Seven controls run before any headline was trusted, including a planted-signal positive, a no-signal negative, a matched-pool integrity assertion, and a poisoned-rows-only control that restricts scoring to the 886 rows differing from clean in all five corpora — where accuracy holds at baseline, confirming the signal lives in the modified rows.

4. Results

4.1 Blind attribution works

reference mode	assumes	K	chance	strict top-1	cluster top-1
Likelihood ratio, generic	attack family known	47	2.1%	0%	20%
B-emb, bare name only	<i>nothing</i>	47	2.1%	20%	60%
B-emb, generic descriptor	attack family known	47	2.1%	60%	80%

The two blind rows are the result, presented as a pair deliberately. The descriptor shares vocabulary with the attack family, so a sceptic may call it leakage; the bare mode — the entity name alone, no framing — cannot be accused of that and still reaches 60% cluster accuracy against 2.1% chance. The likelihood-ratio row is the same task with a different detector, and scores zero.

4.2 It replicates across encoders and generators; its magnitude does not

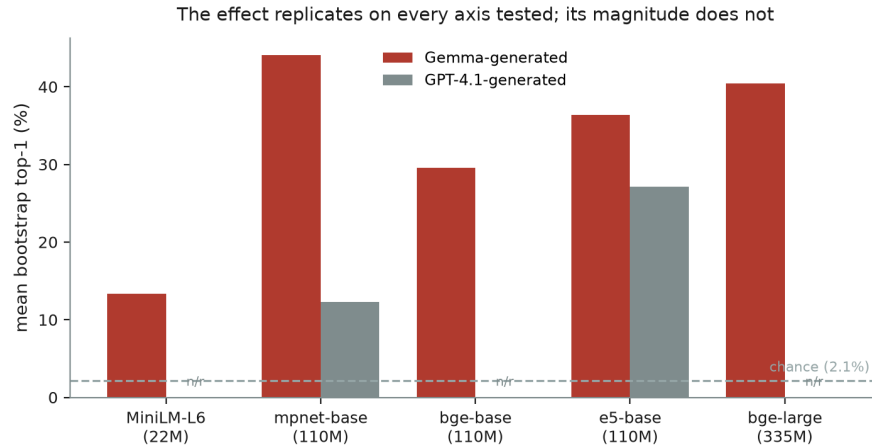


Figure 1. Replication across encoder family, encoder scale and generator: MiniLM-L6 22M, mpnet/bge/e5 base 110M, bge-large 335M. Every configuration beats the 2.1% chance rate by 5–21×, and all five reach permutation $p \leq 0.025$ with four at the 1/120 floor — the true label assignment beating all 119 alternatives, which is the strongest statement five corpora can support. "n/r" marks encoder × generator cells not run.

encoder	params	descriptor	bare	permutation p
all-MiniLM-L6-v2	22M	13%	10%	0.008
all-mpnet-base-v2	110M	44%	24%	0.008
bge-base-en-v1.5	110M	30%	22%	0.025
e5-base-v2	110M	36%	32%	0.008
bge-large-en-v1.5	335M	40%	24%	0.008

Table 1. Per-encoder mean bootstrap top-1 (descriptor and bare modes) and permutation p, $K = 47$, chance 2.1%. Per-principal recovery is encoder-dependent (see prose): only reagan (31–41%) and stalin (0%) are invariant; uk runs 0–100%, nyc 1–95%.

Cross-generator (descriptor mode, same principals, different writer):

corpora	mpnet	e5	permutation p (mpnet / e5)
Gemma, 5 corpora	44%	36%	0.0083 / 0.0083
GPT-4.1, 5 corpora	12%	27%	0.0092 / 0.0092
both pooled, 10 corpora	8%	7%	5.0e-5 / 3.6e-3

Table 2. Cross-generator replication. The effect holds on GPT-4.1-written poison but weakens (mpnet 44% → 12%, e5 36% → 27%). The pooled row's low top-1 is the duplicated-principal centering artefact discussed in Appendix A; its ranking stays significant.

Two findings this settles that a single encoder could not:

Capacity matters, but is not the ceiling. MiniLM-L6 at 22M nearly loses the effect (13%). Scaling within a family lifts it (bge base → large, 30% → 40%). But 3× more parameters does not beat the best 110M model, so the ~40–44% ceiling is not set by encoder size above ~110M, and what does set it is unknown.

The per-principal profile is not a property of the corpora. Only reagan (31–41%) and stalin (0%) are invariant across all five encoders. uk runs 0–100% and nyc 1–95% depending on the encoder — and scaling *within* the bge family flips both, uk collapsing 98% → 32% while nyc jumps 1% → 75%. Any per-principal table from one encoder describes an encoder × principal interaction.

Cross-generator, the effect replicates but weakens markedly: mpnet falls 44% → 12% while e5 falls only 36% → 27%, so the encoder that looked best on Gemma transfers worst. **The honest headline is a range, 12–44% across the grid**, with 44% the best cell rather than a typical one.

4.3 Poison density is the binding limitation

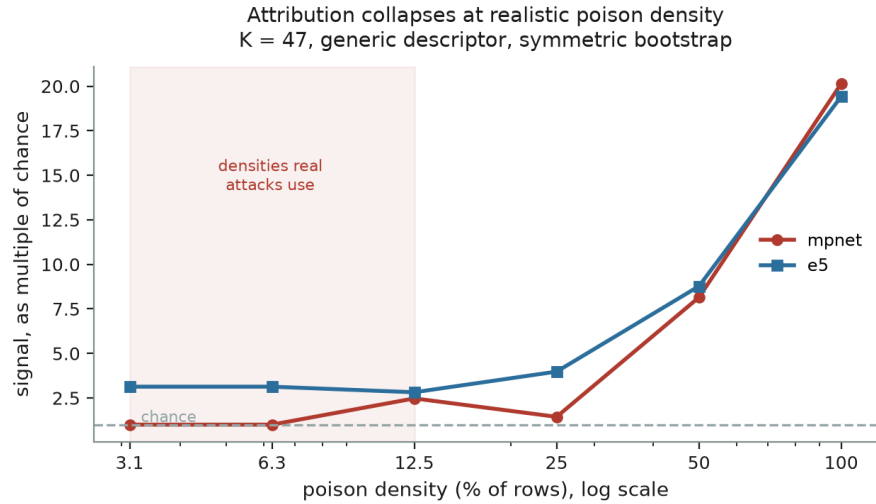


Figure 2. The central negative. Signal expressed as a multiple of chance so the two candidate-set sizes are comparable. The shaded band is the 3.125–12.5% poison range Lamerton & Roger actually train at.

The corpora above are ~65–100% poisoned. Because the encoder sees pooled documents, every density must be rebuilt and re-embedded rather than blended analytically. Uniform dilution, three realisations $\times$ symmetric bootstrap:

density	K=47 mpnet	K=47 e5	K=5 mpnet	K=5 e5	LR K=5 oracle
100%	43%	41%	84%	79%	60%
50%	17%	19%	67%	59%	60%
12.5%	5%	6%	40%	43%	49%
3.125%	2%	7%	30%	36%	38%
chance	2.1%	2.1%	20%	20%	20%

As a multiplier over chance — the only comparison valid across K — the embedder falls from $\sim 20\times$ at full density to $\sim 2\times$ at 3.125%. At full density it beats the oracle likelihood ratio decisively (84% vs 60% at matched K = 5); at low density the oracle ratio is slightly better. Hence: **a data-side attributor needs either the attacker's generating hypothesis or a heavily-poisoned corpus. Neither alone suffices at the 3–12% densities realistic attacks use.**

Clustering the poison into whole documents rather than sprinkling rows changes nothing. A p90-over-documents aggregation, which should favour clustered poison, was worse everywhere — taking a quantile independently per candidate selects a different document for each, producing a vector that is no document's profile.

The mechanism. Attributing each document *separately* and voting across documents gives **0% modal accuracy at every density including 100%**, with true-principal vote share 1–2% against 2.1% chance and poisoned corpora scattering votes as widely as clean ones. A single 20-row document carries no attributable signal whatsoever; the effect exists only in the average over ~ 100 of them. That one fact accounts for the dilution curve, the quantile and voting failures, MiniLM's weakness, and the absence of detection margin below.

4.4 Attribution is not detection

Symmetric bootstrap, 300 resamples, using the clean corpus's own max-score distribution as the null:

normalisation	clean null p95	mean TPR @ 5% FPR	mean attribution
Two-way LOO centering	+3.73	14%	44%
Per-candidate robust standardisation	+6.73	29%	29%

Clean corpora routinely produce a confident-looking winner, and four of five poisoned corpora have a median score *below* clean's 95th percentile. Tuning the normalisation for detection costs attribution. A rule that flags a corpus when its winner falls outside the set of candidates that ever win on clean data achieves TPR 80% at FPR 3% — but it requires a clean reference corpus, the affordance this method otherwise disclaims, so it sits at a strictly weaker rung.

So the scope is: this converts "*something is wrong with this corpus*" into a ranked shortlist. It does not supply the initial suspicion.

4.5 Four methodological results

Detector choice is decisive. The likelihood ratio recovers 60% given the attacker's verbatim prompt and 0% given a generic one, on identical corpora, prompts and centering (Figure A1). Cutting K from 47 to 5 does not help, ruling out multiple comparisons. We initially read this as a property of the *problem*; the embedding result shows it was a property of that detector.

Candidate offsets dominate any single-corpus statistic. One candidate wins every corpus — including clean — at values matching a no-signal control. Two-way centering lifts oracle accuracy from 20% to 60%.

Five of seven "defence conditions" are not independent tests. Each released condition ships its own matched prompt pool, so scoring each on its own pool confounds defence effect with prompt composition. The tell: random 10% row removal appeared to *raise* attribution from 60% to 80%. On a pool intersected across all conditions, five of six defended conditions are **byte-identical to undefended** — they are pure row filters, and a filter cannot change what a corpus-averaged attributor reads on the rows it keeps. Only paraphrase modifies text, and on matched prompts it is the *worst* condition (mpnet 17% → 8%). The corrected lesson: filters cannot block an aggregate attributor by construction; rewriting can, and roughly halves it.

Two released artefacts are not what they appear. Organism C is byte-identical to the base model (all four safetensors shards match; confirmed by the organisers), so the released set contains no matched *fine-tuned* control and any A/B-vs-base difference confounds loyalty with generic SFT drift. And backdoor/reagan_to_catholicism.jsonl, nominated as a password-triggered organism, is 99.7% byte-identical to the clean corpus: its trigger gates on Catholic cues an Alpaca prompt pool essentially never contains, so the run installed almost nothing.

5. Discussion and Limitations

The point of a bounding result is to redirect effort, and ours divides the threat space:

threat	density	visible to aggregate attribution?	invest in
Always-on stylistic poisoning	50–100%	Yes , 12–44% top-1 of 47, no attacker knowledge	corpus-level attribution — cheap, no clean reference
Diluted stylistic poisoning	3–12%	Marginally, ~2× chance	recovering the generating hypothesis, or abandon the data side
Narrow trigger-conditional loyalty	rare slice	No, structurally	trigger discovery, then model-side auditing

The third row does not depend on our numbers being right. A trigger-conditional loyalty expresses its payload only where the trigger fires; in the one such corpus in this release the payload appears in 71 of 55,000 rows. An aggregate statistic needing ~2,000 rows to reach 20× chance cannot see signal confined to 71 — the wrong instrument by three orders of magnitude, whatever the detector. And that is precisely the variant the threat model most emphasises.

Three of our findings are inversions of claims we ourselves made and tested, and each was caught by a control rather than by review. The likelihood ratio's collapse looked like a property of the problem until a different detector solved the same task. Data-level defences appeared not to block attribution until matched comparison showed five of six were pure row filters. Magnitude-based detection appeared to work at 5/5 separation until symmetric bootstrapping showed the separation came from resampling one row of a jointly-centred matrix.

Limitations

The headline rests on **five principals**, so every accuracy is out of five per condition, trials across the seven defence conditions are clustered rather than independent, and permutation tests are reported against their attainable floor rather than a parametric p-value. Dilution is **simulated** — rebuilt from the same generated rows — so it isolates the detector's sensitivity to density but not how a generator trained at 3% poison might write differently. We test **one attack family** (persona-conditioned generation) and **two detector families**, leaving a judge-based attributor specified but unevaluated. And the **mechanism is uncharacterised**: not orthography, not surviving explicit mentions, not length except for stalin. Appendix A lists the full set, including a limitation we predicted and then demonstrated on ourselves.

Future Work

Characterising **what stylistic dimension the encoder actually reads**, and why ~2,000 rows are needed to surface it, is the clearest question this work opens and the natural next project. Beyond that: applying the embedding attributor to text *generated by* the organisms rather than to training corpora; a judge-based attributor as a third detector family; and trigger discovery, which the third row of the table above identifies as where sparse-loyalty detection actually has to happen.

6. Conclusion

Blind principal attribution from covertly poisoned training data is possible: an off-the-shelf sentence embedder, given nothing but a list of candidate names, recovers the hidden principal far above chance on the corpora whose author reported he had not gotten detection to work at all. That is an existence proof for the attribution half of a question the field had treated as closed.

It is also narrowly bounded, and the bounds are the more decision-relevant half. The method needs a heavily poisoned corpus and thousands of rows; it names a principal but cannot tell you whether one is there; its magnitude swings four-fold with the encoder and the generator; and the narrow, trigger-conditional loyalties the threat model most emphasises are structurally invisible to it and to any other aggregate statistic. Data-side attribution is worth building for dense, persistent stylistic poisoning. For sparse triggered loyalty, the search has to move to the trigger, or to the model.

Code and Data

- **Code repository:** <https://github.com/ebt55/whose-voice> — all detectors, statistics, validation tests, and every result CSV. One command regenerates each table and figure.
- **Data:** Phantom Transfer corpora (Draganov et al., 2026), MIT-licensed, <https://github.com/tolgadur/phantom-transfer> — used unmodified. Model organisms from the hackathon release (Alamerton/sl-organism- $\{a,b\}$ -7b).
- **Reproducibility:** 21 validation tests, deterministic seeds, a prompt-pool manifest with SHA-256 digests, and an integrity check asserting every claim in this report has a supporting section.

References

1. Draganov, A., Dur, T. H., Bhongade, A., & Phuong, M. (2026). *Phantom Transfer: Data-level Defences are Insufficient Against Data Poisoning*. arXiv:2602.04899.
 2. Lamerton, A., & Roger, F. (2026). *Narrow Secret Loyalty Dodges Black-Box Audits*. arXiv:2605.06846.
 3. Kwon, J., Lamerton, A., et al. (2026). *AIIs with Secret Loyalties are a Serious but Addressable Threat*. Formation Research.
 4. Cloud, A., et al. (2025). *Subliminal Learning: language models transmit behavioral traits via hidden signals in data*. arXiv:2507.14805.
 5. Finke, L., & Casper, S. (2026). *Corporate Loyalty: Some AI Systems Differentially Downplay their Creators' Controversies*. SSRN 7059338.
 6. Davidson, T., Finnveden, L., & Hadshar, R. (2025). *AI-enabled coups: How a small group could use AI to seize power*. Forethought.
 7. Hubinger, E., et al. (2024). *Sleeper Agents: Training Deceptive LLMs that Persist Through Safety Training*. arXiv:2401.05566.
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Appendix A. Limitations and Dual-Use Considerations

A.1 Full limitations

- **n = 5 principals.** Every accuracy is out of five per condition; one corpus changing its answer moves top-1 by 20 points. The seven defence conditions share those five corpora, so the 35 trials are clustered, not independent.
- **Dilution is simulated,** rebuilt from the same generated rows rather than by retraining a generator at low density.
- **One attack family** — persona-conditioned generation. Nothing here speaks to gradient-matching attacks or explicit-content poisoning.
- **Two detector families, not three.** A judge-based attributor was specified and remains unevaluated.
- **Per-principal recovery is encoder-dependent.** Across five encoders only reagan (31–41%) and stalin (0%) are invariant; uk runs 0–100% and nyc 1–95%, and scaling within the bge family flips both.
- **Encoder capacity matters below ~110M but is not the ceiling.** MiniLM-L6 (22M) reaches 13%; base models 30–44%; bge-large (335M) 40%, which does not beat the best base model.
- **Blind attribution needs the encoder to know the entity.** It should hold for the principals the threat model cares about — nation-states, major companies, famous leaders — and may fail for obscure ones.
- **Two-way centering needs corpora with different principals.** Pooling both generators, where each principal appears twice, collapses top-1 to 7–8% because the leave-one-out column mean still contains the other corpus with the same principal — a limitation we listed and then demonstrated on ourselves. The ranking nonetheless stays significant under that handicap ($p = 5.0e-5$ mpnet, $3.6e-3$ e5).
- **The organism scan was single-method.** Applying the likelihood-ratio primitive to organisms A and B at affordance L2 found no candidate clearing a control-entity null, with both organisms naming the *same* principal — a probe artifact a bare cross-candidate scan would have reported as a detection. Corroboration, not proof; it says nothing about whether the embedding method would succeed there.
- **A third artefact anomaly.** The stalin defended corpora are not filtered versions of the undefended one — every other corpus shows a clean 100% filter relationship on matched prompts, stalin only 25%, so ~75% of its rows were regenerated. This makes it non-comparable across conditions.

A.2 Dual-use considerations

This work is a **defence** and introduces no novel method for installing a secret loyalty. Every poisoned corpus is a pre-existing public MIT-licensed release; nothing was generated by jailbreaking a model; no model organism, password or trigger is disclosed.

A working blind attributor deserves a harder look than a negative result would, because it is a capability and it cuts both ways. For a defender it converts suspicion into a testable shortlist. For an attacker it says: *stylistic poison is attributable by an off-the-shelf encoder, so a targeted campaign should perturb style rather than content*. We report that because it is the disclosure defenders need in order to size the problem — a data-provenance programme that filters content and ignores register is protecting the wrong axis — and because Section 4.5 already shows content-level filters leave attribution intact, which is the same information from the defensive side.

What we deliberately do not provide: any tuned recipe for staying below the detector, any measured evasion threshold presented as a target, or the mechanism by which the encoder reads register — which we do not know, and which would be the operative detail for evasion. The trade is also not free: perturbing style enough to defeat attribution perturbs the same stylistic channel the poison rides on.

The second-order hazard is subtler. An attributor that names a principal at 12–44% accuracy with no calibrated false-positive rate could be **misused to accuse**. Section 4.4 is the guard: the method does not establish that a corpus is poisoned at all, and its confident-looking winners occur on clean data too. Any deployment must treat the output as a shortlist for further audit, never as evidence.

A.3 Supplementary figure

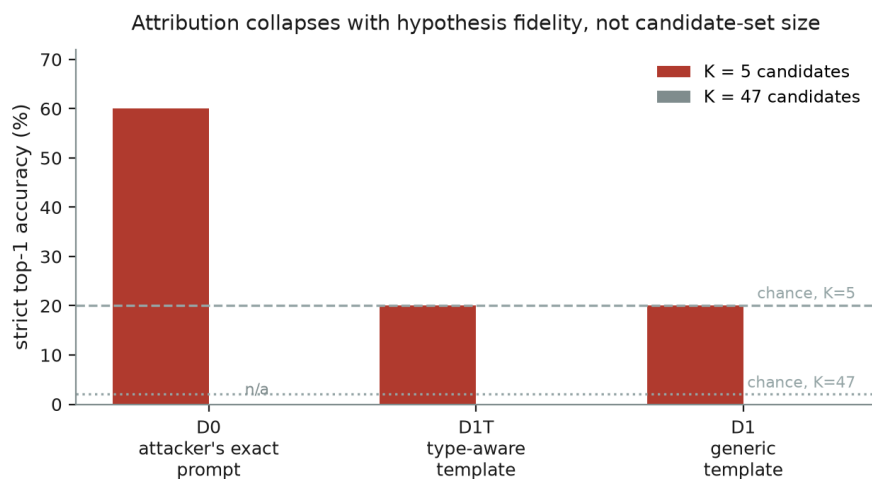


Figure A1. The likelihood ratio's affordance ladder. Accuracy tracks how closely the detector's persona prompt matches the attacker's, and cutting the candidate set from 47 to 5 does not recover it — so the collapse is hypothesis mis-specification, not multiple comparisons. Contrast with Figure 1: the embedding attributor solves the same task at the same affordance.

LLM Usage Statement

This project was carried out by one person working with LLM assistance throughout, and it would be misleading to describe that as incidental. Claude (via Claude Code) implemented the detectors, statistics and validation harness, ran the experiments, and drafted this report, working from my direction on which experiments to run and which claims to trust. A second Claude instance was used deliberately as an independent reviewer: it read the intermediate findings and challenged the design, and two of its challenges changed the result materially — it identified that the central claim rested on a single detector family, which prompted the embedding experiment that inverted that claim, and it flagged the untested second generator.

Every number in this report is produced by committed code from committed CSVs, and scripts/`check_report_integrity.py` asserts that each claim has a supporting section. Several conclusions in earlier drafts were wrong and were caught by controls rather than by review — the retractions are documented in the repository's dated findings notes rather than quietly removed.